Daria Florea

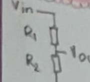
CHEATSHEET

Cheatsheet EEA

Ordime de mărime

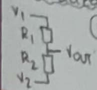
Multiplu	Pref.	Simb.
^15	Peta	P
^12	Terra	T
^9	Giga	G
^6	Mega	M
^3	Kilo	K
^-3	Mili	m
^-6	Micro	µ
^-9	nano	n
^-12	pico	p
^-15	femto	f

Divizor de tensiune



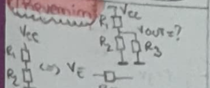
$V_{out} = V_{in} \cdot \frac{R_2}{R_1+R_2}$

Generalizare

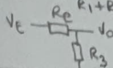


$\boxed{V_{out} - V_2 = (V_1 - V_2) \cdot \frac{R_2}{R_1+R_2}}$

Thevenin



$V_E = V_{cc} \cdot \frac{R_2}{R_1+R_2}$ $R_E = R_1 \| R_2$



→ div. de tens.

$[C]_{SI} = F$ (Farad)

→ prop. primară → capacitatea

$[L]_{SI} = H$ (Henry)

→ -||- → inductanța

Imp. de ieșire surs. de tensiune ideală = 0

Dioda

- → joncțiune PN (Siliciu)
- → substrat de tip Si

Tensiune de prag - 0,6V - 1V

Curent direct max - 10mA - 100A

Ten. inv. max. - 1000V

Curent de saturație - 10 - 100 nA

Timp de comutație - 10ns - 10µs

Putere disipată 100 mW - 100W

Dioda Schottky

- → joncțiune metal-semiconductor
- → substrat de Si tip P

Ten. de prag - 0,3V

Cur. direct max. - 10mA - 100A

Ten. inv. max. - 30-60V

Cur. de sat. - 100nA - 1mA

Timp de comutație - 100ps - 100ns

Putere disipată - 100mW - 100W

Avantaje

- cădere de ten. redusă
- vit. mare de comutare

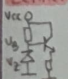
Dezavantaje

- tens. inv. redusă
- cur. de sat. mare

LED (light - emitting diode)

- → diodă semiconductoare ce emite lumină la pol. dir. a joncțiunii p-n
- → cădere de tensiune 1,6V - 4V

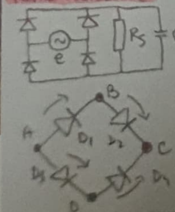
Zenner



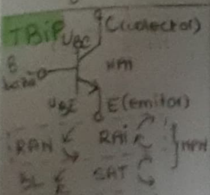
$V_{cc} > V_Z$ ⇒ $U_B = V_Z$, diodă deschisă

$V_{cc} < V_Z$ ⇒ diodă închisă, $U_B = V_{cc}$

Punte redresoare



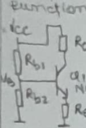
TBIP



$i_C = \beta i_B + \underbrace{I_{CEO}}_{\text{neglijat de obicei}}$

[20,500]

PSF = punct static de funcționare



$U_B = V_{cc} \cdot \frac{R_{b2}}{R_{b1}+R_{b2}}$

$U_E = U_B - U_{BE}$

$U_E = U_{RE} = I_E \cdot R_E$

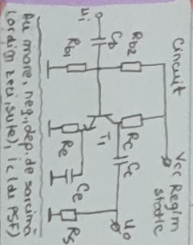
$\Rightarrow i_E = \frac{U_E}{R_e}$ $i_E = i_C$

$U_{CE} = V_{CC} - U_{R_C} - U_{R_e}$

$U_{R_C} = i_C \cdot R_C$

Conexiuni fundamentale

EC (emitor comun)

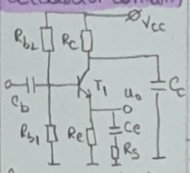


Au mare, neg., dep. de sarcină (ordin zeci, sute), ic (de PSF)

- Ai → mare
- Ap → ft. mare
- Zi mediu (prea mic) (sute Ω/KΩ)
- Zo mediu (prea mare)

DOAR INTERMEDIAR

CC (colector comun)



Au - subunitară (1)

Ai - mare

Ap - medie

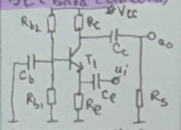
Zi - mare (zeci de KΩ)

Zo - mic

(bun pt. intrare/ieșire)

- repetorul pe emitor este un ft. bun etaj de adaptare
- repetor de tensiune → conexiune CC

BC (bază comună)



- → Au mare, pozitivă, depinde de ic (PSF)
- → Ai → (-)1
- → Zi - mică (zeci de Ω)
- → Zo - mare
- → bun pentru intrare în curent
- → tranzistorul se comp. la ieșire ca un gen. de curent și în regim dinamic
- → sarcini dinamice/generatoare de curent constant

AMPLIFICATOARE

Amplificator ideal

$Z_i \to +\infty$ $A_u \to +\infty$

$Z_o \to 0$ A_0 - inf

Model general

$Z_i = \frac{u_i}{i_i}$, pt. $u_o' = 0$

$A_u = \frac{u_o}{u_i}$, gol

$A_u' = \frac{u_o'}{u_i}$, în sarcină

$u_i = u_g \frac{Z_i}{Z_i + Z_g}$, vrem $Z_i \gg Z_g$

$u_o' = u_o \cdot \frac{Z_s}{Z_s + Z_o}$, vrem $Z_o \ll Z_s$

$A_u' = A_u \cdot \frac{Z_s}{Z_s + Z_o}$

Rail-to-rail output = AO tensiunea de ieșire poate fi app sau chiar egală cu tensiunea de alim. (V+ sau V-)

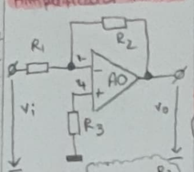
Pt. AO valoarea tensiunii de decalaj (offset voltage) - sute de micro V

AO ideal - curenții de intrare = 0

Reacție negativă

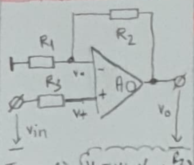
$v_+ = v_-$ $i_+ = i_- = 0$

Amplificator inversor



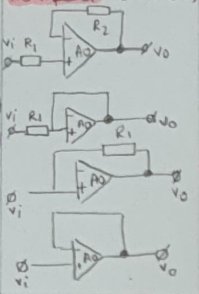
Formulă: $\boxed{V_0 = -v_i \cdot \frac{R_2}{R_1}}$

Amplificator neinversor

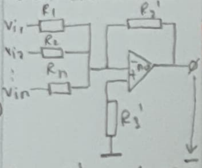


Formulă: $\boxed{V_0 = v_{in} \cdot \left(1 + \frac{R_2}{R_1}\right)}$

AO repetor $v_0 = v_i$ $(A_v = 1)$

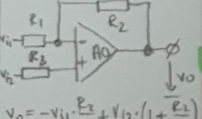


AO sumator inversor



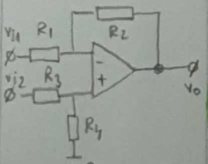
$v_0 = -R_2' \cdot \text{sum}(v_{ij}/R_j)$, $j = \overline{1,n}$

AO inversor-neinv.



$v_0 = -v_{i1} \cdot \frac{R_2}{R_1} + v_{i2} \cdot \left(1 + \frac{R_2}{R_1}\right)$

Amplificator diferențial cu AO

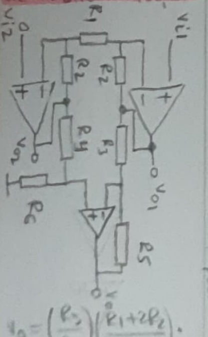


$v_0 = v_{i2} \cdot \frac{R_4}{R_3 + R_4} \cdot \left(1 + \frac{R_2}{R_1}\right) - v_{i1} \cdot \frac{R_2}{R_1}$

Dacă $R_2/R_1 = R_3/R_4 \Rightarrow$

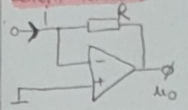
$\boxed{v_0 = \frac{R_2}{R_1}(v_{i2} - v_{i1})}$

INA (amplificator de instrumentație)



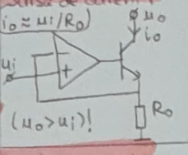
$v_0 = \left(\frac{R_5}{R_3}\right)\left(\frac{R_1 + 2R_2}{R_1}\right) \cdot (v_{i2} - v_{i1})$

Convertor curent-tensiune $\boxed{U_0 = -iR}$



Sursă de curent 1

$\boxed{i_0 \approx u_i/R_0}$

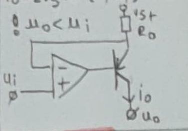


$(u_0 > u_i)$!

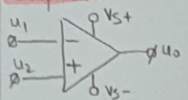
Sursă de curent 2

$i_0 = [V_S + - (-) u_i]/R_0$

! $u_0 < u_i$



Comparator



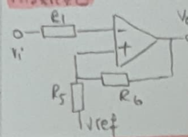
$u_1 > u_2 \Rightarrow u_0 = V_{S-}$

$u_2 > u_1 \Rightarrow u_0 = V_{S+}$

Comparator cu histerezis

$V_+ > V_i$

$V_+ < V_i$



Tensiunea de decalaj Ud (offset voltage)

+Ud (se adună)

Ud se conectează la +-ul AO.

Neinversor

$u_0 = \left(1 + \frac{R_2}{R_1}\right)(u_i + U_d)$

Inversor

$u_0 = -u_i \cdot \frac{R_2}{R_1} + U_d(1 + R_2/R_1)$

Amplificator dif.

Rejecția modului comun/Factorul de rejecție al modului comun → (CMRR)

$CMRR = 10\log_{10}\left(\frac{A_d}{A_c}\right)^2 = 20\log_{10}\left(\frac{A_d}{|A_c|}\right) dB$

FILTRE

FTJ/FTS → frec. de tăiere este frecv. la care puterea semnalului de ieșire scade, față de puterea în banda de trecere, la (50%)

Banda de trecere, punctul de -3dB

- → punctele unde amplificarea de putere scade la jum.
- → tensiunea scade de rad. din 2 ori
- → punct de -3dB

Pt. filtru de ord 1

→ panta -20dB/dec.

ord 2 → -40 dB/dec.

* tensiunea este la 70,7% din val. max din banda de trecere

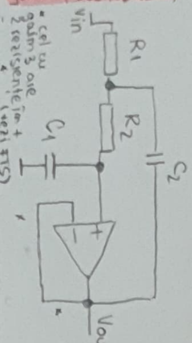
Extra !! $\omega = 2\pi f$

$X_C = \frac{1}{2\pi f C}$

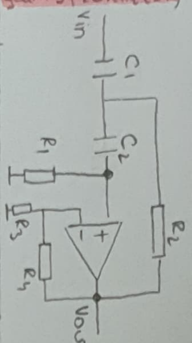
Conductanța $G = \frac{1}{R}$

FTJ Activ de ord 2, gain 1, 200Hz (SK)

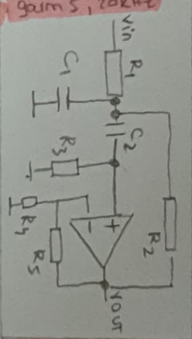
* cel cu gain 3 are 2 rezistențe în + (vezi FTS)



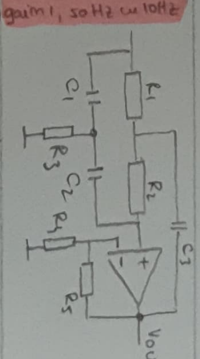
FTS Activ de ord 2, gain 3, 20kHz (SK)



FTB Activ de ord 2, gain 5, 20kHz

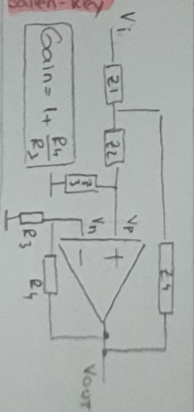


FRB Activ de ord 2, gain 1, 50Hz cu 10Hz

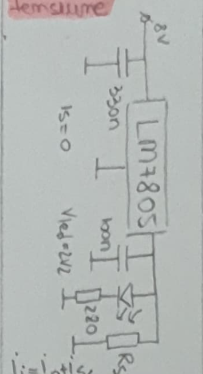


Sallen-key

$\boxed{Gain = 1 + \frac{R_4}{R_3}}$



Stabilizator de tensiune



$i_i = i_o + i_s$

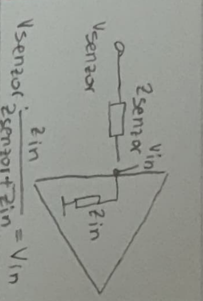
$i_1 = i_o + i_s = i_o$

P_u = putere utilă = $i_{out} \cdot V_{out}$

P_c = putere consumată = $i_{in} \cdot V_{in}$

$\eta = P_u / P_c$

Amplificator neideal



$V_{senzor} \cdot \frac{Z_{in}}{Z_{senzor} + Z_{in}} = V_{in}$